

MODULE 1: Introduction to Network Simulator and Sniffing Tools

1. What is a Computer Network?

A computer network is a collection of interconnected devices such as computers, servers, printers, and mobile devices that communicate and share resources through communication links.

Advantages:

- Resource sharing
- File sharing
- Internet access
- Communication

Example:

College computer lab connected through LAN.

2. What is NS-3?

NS-3 (Network Simulator 3) is an open-source network simulation software used to study and analyze network behavior without creating a physical network.

Features:

- Supports wired and wireless networks
- Open source
- Supports C++ and Python
- Used in research and education

Uses:

- Performance testing
 - Protocol analysis
 - Network design
-

3. Why do we use NS-3?

NS-3 helps simulate real network scenarios before actual deployment.

Benefits:

- Saves cost
 - Saves time
 - No hardware required
 - Easy experimentation
-

4. What is NetAnim?

NetAnim is a graphical animation tool used with NS-3.

Purpose:

It visually shows:

- Nodes
- Data transmission
- Packet movement

Benefit:

Makes simulation easy to understand.

5. What is PyViz?

PyViz is a Python-based visualization tool for NS-3.

Functions:

- Shows network nodes
 - Displays packet flow
 - Visual debugging
-

6. What is Wireshark?

Wireshark is a network protocol analyzer used to capture and inspect packets traveling through a network.

Uses:

- Network troubleshooting
- Security analysis
- Packet inspection

Features:

- GUI-based
 - Real-time packet capture
 - Supports many protocols
-

7. What is Packet Sniffing?

Packet sniffing is the process of capturing and monitoring packets transmitted over a network.

Purpose:

- Analyze traffic
- Detect problems
- Security monitoring

Tools:

- Wireshark
 - Tcpdump
-

8. What is Tcpdump?

Tcpdump is a command-line network packet analyzer.

Advantages:

- Lightweight
- Fast
- Suitable for Linux servers

Example:

```
tcpdump -i eth0
```

Captures packets from interface eth0.

9. Difference Between Wireshark and Tcpdump

Wireshark

GUI Based

User Friendly

Detailed Visualization

Easy Packet Analysis

Tcpdump

Command Line

Advanced Users

Fast Capture

Less Visual

Linux Commands Viva**10. What is Linux?**

Linux is an open-source operating system based on Unix.

Features:

- Multi-user
- Multi-tasking
- Secure
- Free

11. What does pwd command do?

pwd

Meaning:

Print Working Directory

Purpose:

Displays current directory path.

12. What does ls command do?

ls

Lists files and folders in current directory.

Common options:

ls -l

Detailed listing.

ls -a

Shows hidden files.

13. What is cd command?

cd foldername

Changes current directory.

14. What is mkdir command?

mkdir test

Creates a new directory named test.

15. What is rm command?

rm file.txt

Deletes a file.

`rm -r folder`

Deletes a directory recursively.

16. What is ping command?

`ping google.com`

Tests connectivity between two devices.

Uses:

- Network troubleshooting
 - Check host availability
-

17. What is ifconfig?

Displays network interface information.

Shows:

- IP Address
 - MAC Address
 - Interface status
-

MODULE 2: Network Topology

18. What is Network Topology?

Network topology refers to the arrangement of devices and communication links in a network.

Types:

- Bus
 - Star
 - Ring
 - Mesh
 - Hybrid
-

19. Explain Bus Topology.

All devices connect to a single communication cable called backbone.

Advantages:

- Low cost
- Easy installation

Disadvantages:

- Backbone failure affects entire network
 - Difficult troubleshooting
-

20. Explain Star Topology.

All devices connect to a central switch or hub.

Advantages:

- Easy maintenance
- Failure of one node doesn't affect others

Disadvantages:

- Switch failure affects network

Most commonly used topology.

21. Explain Ring Topology.

Devices form a circular path.

Advantages:

- Organized communication

Disadvantages:

- One break can stop communication
-

22. Explain Mesh Topology.

Every node is connected to every other node.

Advantages:

- High reliability
- Multiple paths available

Disadvantages:

- Expensive
 - Complex
-

23. Explain Hybrid Topology.

Combination of two or more topologies.

Example:

Star-Bus network.

24. Which topology is most reliable?

Mesh topology because multiple communication paths exist.

MODULE 3: Basic Network Protocol Models

25. What is a Protocol?

A protocol is a set of rules that governs communication between devices.

Examples:

- TCP
 - UDP
 - HTTP
 - FTP
-

26. What is Client-Server Architecture?

A communication model where:

Client:

Requests services.

Server:

Provides services.

Example:

Web browser and web server.

27. What is TCP?

TCP stands for Transmission Control Protocol.

Features:

- Reliable
- Connection-oriented
- Error detection
- Acknowledgement system

Uses:

- Web browsing
- Email
- File transfer

28. What is UDP?

UDP stands for User Datagram Protocol.

Features:

- Fast
- Connectionless
- No acknowledgement

Uses:

- Video streaming
- Online gaming
- VoIP

29. Difference Between TCP and UDP

TCP	UDP
Reliable	Unreliable
Connection Oriented	Connectionless
Slow	Fast
Uses ACK	No ACK

30. What is FTP?

FTP stands for File Transfer Protocol.

Purpose:

Transfer files between client and server.

Port Number:

21

31. What is HTTP?

HTTP stands for HyperText Transfer Protocol.

Purpose:

Transfer web pages.

Port Number:

80

32. What is HTTPS?

HTTPS is secure HTTP using SSL/TLS encryption.

Port Number:

443

Advantage:

Secure communication.

33. What is DHCP?

DHCP stands for Dynamic Host Configuration Protocol.

Function:

Automatically assigns:

- IP address
- Subnet mask
- Gateway
- DNS

Ports:

67 and 68

34. What is DNS?

DNS stands for Domain Name System.

Function:

Converts domain names into IP addresses.

Example:

google.com → 142.x.x.x

Port:

53

35. What is SMTP?

SMTP stands for Simple Mail Transfer Protocol.

Purpose:

Sending emails.

Port:

25

MODULE 4: Network Traffic Analysis

36. What is Network Monitoring?

The process of observing network traffic and performance.

Goals:

- Detect issues

- Improve performance
 - Ensure security
-

37. What is Protocol Analysis?

Examining network packets to understand communication.

Tool:

Wireshark

38. What is ARP?

ARP stands for Address Resolution Protocol.

Purpose:

Converts IP address into MAC address.

Example:

Finding MAC address of a device before sending data.

39. What is MAC Address?

A unique hardware address assigned to a network interface.

Length:

48 bits

Example:

00:1A:2B:3C:4D:5E

40. What is ICMP?

ICMP stands for Internet Control Message Protocol.

Uses:

- Error reporting
- Connectivity testing

Example:

Ping command

41. What is Packet Filtering in Wireshark?

Filtering displays only selected packets.

Example:

http

Shows only HTTP packets.

42. What is DNS Query?

A request sent to DNS server for domain resolution.

MODULE 5: Network Performance Evaluation

43. What is Network Performance?

Measurement of network efficiency and quality.

Parameters:

- Throughput
 - Latency
 - Packet loss
 - Jitter
-

44. What is Throughput?

Actual amount of data transferred successfully per second.

Unit:

Mbps or Gbps

45. What is Bandwidth?

Maximum capacity of a network link.

Example:

100 Mbps network.

46. Difference Between Throughput and Bandwidth

Bandwidth = Maximum capacity

Throughput = Actual achieved speed

47. What is Latency?

Time taken for data to travel from source to destination.

Unit:

Milliseconds (ms)

48. What is Packet Loss?

Packets that fail to reach destination.

Causes:

- Congestion
 - Hardware failure
 - Network errors
-

49. What is Jitter?

Variation in packet arrival time.

Important for:

- Video calls
 - Online gaming
-

MODULE 6: Real Time Network Problem Solving

50. What is Network Troubleshooting?

Process of identifying and resolving network issues.

Steps:

1. Identify problem
2. Analyze cause

3. Implement solution
4. Verify result

51. How do you check connectivity?

Using:

ping

52. How do you check routing information?

route

or

netstat -r

53. How do you check network interfaces?

ifconfig

or

ip addr

54. What is OMNeT++?

An open-source network simulation framework.

55. What is QualNet?

Commercial network simulator for large-scale network analysis.

56. What is NetSim?

Network simulation software used for academic and industrial purposes.

MOST IMPORTANT EXAMINER QUESTIONS**Explain OSI Model.**

1. Physical Layer
2. Data Link Layer
3. Network Layer
4. Transport Layer
5. Session Layer
6. Presentation Layer
7. Application Layer

Explain TCP/IP Model.

1. Network Access Layer
2. Internet Layer
3. Transport Layer
4. Application Layer

Difference Between IP Address and MAC Address.

IP Address	MAC Address
Logical Address	Physical Address
Can Change	Fixed
Layer 3	Layer 2

Which protocol does Ping use?

ICMP

Which protocol assigns IP addresses automatically?

DHCP

Which protocol converts domain names into IP addresses?

DNS

Which topology is most commonly used in companies?

Star Topology

Which topology provides highest fault tolerance?

Mesh Topology

ASSIGNMENT 1 (NS3 Installation)

Why do we install NS-3?

To simulate computer networks without creating physical networks.

Why is g++ installed?

Because NS-3 programs are written mainly in C++.

Why is Python installed?

NS-3 supports Python scripting and PyViz.

Why do we install Wireshark?

To analyze captured packets (.pcap files).

Why is tcpdump installed?

To capture and read packet traces from terminal.

What is a PCAP file?

Packet Capture file containing network traffic data.

Why is NetAnim installed?

To visualize packet movement graphically.

What does ./test.py do?

Tests whether NS-3 installation is successful.

What does

`./waf --run hello-simulator`
do?

Compiles and runs the hello-simulator program.

What is WAF?

WAF is NS-3's build system used to compile and run simulations.

ASSIGNMENT 2

POINT TO POINT TOPOLOGY

What is Point-to-Point topology?

A topology where two devices communicate using a dedicated link.

Why

`nodes.Create(2);`

?

Creates 2 nodes.

Examiner may ask:

What happens if you write

`nodes.Create(5);`

?

Five nodes will be created.

What is NodeContainer?

A container used to store network nodes.

Why

`PointToPointHelper pointToPoint;`

?

Creates a point-to-point communication channel.

Why

`pointToPoint.SetDeviceAttribute("DataRate","5Mbps");`

?

Sets transmission speed to 5 Mbps.

What is DataRate?

Maximum amount of data transferred per second.

What happens if DataRate becomes 10Mbps?

Transmission becomes faster.

Why

`SetChannelAttribute("Delay","2ms");`

?

Adds propagation delay of 2 milliseconds.

What is Propagation Delay?

Time taken by packet to travel from sender to receiver.

Why

`InternetStackHelper stack;`

?

Installs TCP/IP protocol stack.

Why

`stack.Install(nodes);`

?

Adds networking capability to nodes.

Why

Ipv4AddressHelper address;

?

Assigns IP addresses.

Why

address.SetBase("10.1.1.0","255.255.255.0");

?

Creates subnet 10.1.1.0/24.

Why use 255.255.255.0?

Subnet mask defining network and host portions.

What is

UdpEchoServerHelper

?

Creates a UDP Echo Server.

What is Echo Server?

Receives packet and sends same packet back.

Why Port 9?

Port 9 is traditionally used for Echo Service.

Why

serverApps.Start(Seconds(1.0));

?

Server starts at 1 second.

Why client starts at 2 seconds?

Server must start before client sends packets.

What happens if client starts first?

Packets may be lost because server isn't running.

Why

PacketSize = 1024

?

Each packet contains 1024 bytes.

Why

MaxPackets = 1

?

Client sends only one packet.

Why use UDP?

Simple communication with low overhead.

Why

MobilityHelper mobility;

?

Defines node movement or positions.

Why

ConstantPositionMobilityModel

?

Nodes remain stationary.

What is NetAnim XML file?

Stores animation data for visualization.

Why

AnimationInterface anim("first.xml");

?

Creates animation output file.

Why

EnablePcapAll("first");

?

Captures packets for Wireshark analysis.

What does Simulator::Run() do?

Starts simulation execution.

What does Simulator::Destroy() do?

Releases allocated resources.

BUS TOPOLOGY PROGRAM

How many nodes are created?

nNodes = 5

Five nodes.

Why is node 0 special?

Acts as central node connected to all others.

Why for loop starts from 1?

Node 0 is already used as central node.

What is

NodeContainer pair

?

Creates connection between two nodes.

Why

10.1.1.0

used?

Creates separate subnet for every connection.

Why use

PacketSinkHelper

?

Receives packets sent by clients.

What is Packet Sink?

Application that receives incoming packets.

What is

OnOffHelper

?

Traffic generator that alternates ON and OFF states.

Why

UdpSocketFactory

?

Uses UDP protocol.

What is

PopulateRoutingTables()

?

Automatically creates routing tables.

Why assign node positions?

To display topology correctly in NetAnim.

ASSIGNMENT 3

UDP CLIENT SERVER

What is UDP?

User Datagram Protocol.

Connectionless transport protocol.

Why is UDP called connectionless?

No connection establishment before communication.

Does UDP guarantee delivery?

No.

Why use UDP instead of TCP?

Lower overhead and faster communication.

Applications of UDP?

- DNS
 - Video Streaming
 - Gaming
 - VoIP
-

Why

nodes.Create(2);

?

One client and one server.

Why use CSMA?

To simulate Ethernet LAN.

What is CSMA?

Carrier Sense Multiple Access.

Devices check channel before transmission.

Why

5Mbps

used?

Defines Ethernet speed.

Why

Delay = 2ms

?

Defines network latency.

Why

MTU = 1400

?

Maximum Transmission Unit size.

What is MTU?

Maximum packet size transferable without fragmentation.

Why enable PCAP?

Generate capture file for Wireshark.

Why

10.1.1.0

?

Network address.

What is

serverAddress

?

Stores destination IP of server.

Why port 4000?

Communication endpoint for client-server application.

What is a Port Number?

Logical communication endpoint.

Why

maxPacketSize = 1024

?

Each UDP packet contains 1024 bytes.

Why

maxPacketCount = 320

?

Client sends 320 packets.

What is

interPacketInterval = 0.05

?

50 ms delay between packets.

Why use Flow Monitor?

Measures network performance.

What statistics does Flow Monitor provide?

- Throughput
- Delay

- Packet Loss
- Sent Packets
- Received Packets

What is Throughput?

Actual successful data transfer rate.

How is throughput calculated?

Received bits ÷ transmission time.

Why serialize Flow Monitor results?

To save statistics in XML format.

VERY DANGEROUS EXAMINER QUESTIONS

Difference between Node and NodeContainer?

Node = Single device

NodeContainer = Collection of nodes

Difference between DataRate and Delay?

DataRate = Speed

Delay = Travel time

Difference between TCP and UDP?

TCP	UDP
Reliable	Unreliable
Connection Oriented	Connectionless
ACK Used	No ACK
Slow	Fast

Difference between NetAnim and Wireshark?

NetAnim → Visualization

Wireshark → Packet Analysis

Difference between PCAP and XML?

PCAP → Packet Capture

XML → NetAnim Visualization

Why do we use Simulator::Run() and Simulator::Destroy()?

Run executes simulation.

Destroy frees memory after completion.

Most Important Linux Commands for Viva

1. pwd

Command

pwd

Full Form

Print Working Directory

Use

Displays current directory path.

2. ls

Command

ls

Use

Lists files and folders.

Important Options

ls -l

Detailed information.

ls -a

Shows hidden files.

ls -la

Detailed + hidden files.

3. cd

Command

cd foldername

Use

Change directory.

Examples

cd Desktop

cd ..

Moves one directory back.

4. mkdir

Command

mkdir test

Use

Creates new directory.

5. rmdir

Command

rmdir test

Use

Deletes empty directory.

6. rm

Command

rm file.txt

Use

Deletes file.

Delete folder

rm -r foldername

7. cp

Command

cp file1.txt file2.txt

Use

Copies file.

8. mv

Command

mv old.txt new.txt

Use

Move or rename files.

9. cat

Command

cat file.txt

Use

Displays file contents.

10. touch

Command

touch file.txt

Use

Creates empty file.

11. clear

Command

clear

Use

Clears terminal screen.

Networking Commands

12. ifconfig

Command

ifconfig

Use

Displays network interface details.

Shows

- IP Address
 - MAC Address
 - Network Status
-

13. ip addr**Command**

ip addr

Use

Modern replacement for ifconfig.

14. ping**Command**

ping google.com

Use

Checks network connectivity.

Protocol Used

ICMP

Viva Question

Which protocol does ping use?

Answer: ICMP

15. traceroute**Command**

traceroute google.com

Use

Shows path taken by packets.

16. netstat**Command**

netstat

Use

Displays network connections.

Routing Table

netstat -r

17. route**Command**

route

Use

Displays routing table.

18. arp

Command

arp -a

Use

Displays ARP table.

Viva Question

What is ARP?

Address Resolution Protocol.

Converts IP address to MAC address.

19. hostname

Command

hostname

Use

Displays system name.

20. nslookup

Command

nslookup google.com

Use

Finds IP address of domain.

Viva Question

Which protocol helps domain name resolution?

DNS

Wireshark Commands / Filters

Show HTTP packets

http

Show TCP packets

tcp

Show UDP packets

udp

Show DNS packets

dns

Show ARP packets

arp

Show ICMP packets

icmp

Show packets from specific IP

ip.addr == 10.1.1.1

Tcpdump Commands

Capture packets

tcpdump

Capture on specific interface

tcpdump -i eth0

Save capture

tcpdump -w capture.pcap

Read capture file

tcpdump -r capture.pcap

NS-3 Commands

Build NS-3

./build.py --enable-examples --enable-tests

Use

Compiles NS-3.

Test Installation

./test.py

Use

Checks whether NS-3 installed correctly.

Run Program

./waf --run hello-simulator

Use

Runs simulation.

Run Custom Program

./waf --run scratch/first

Use

Runs first.cc program.

NetAnim Commands

Open NetAnim

./NetAnim

Use

Launch NetAnim visualizer.

Commands Most Asked in Viva (Learn These First)

Command	Use
pwd	Current directory
ls	List files
cd	Change directory
mkdir	Create folder
rm	Delete file
cat	Display file
ping	Check connectivity
ifconfig	Show IP details
ip addr	Show network interfaces
arp -a	Show ARP table
netstat -r	Show routing table
nslookup	DNS lookup
tcpdump	Packet capture
./test.py	Test NS3
./waf --run hello-simulator	Run NS3 example
./NetAnim	Open NetAnim

What is a Protocol?

A **protocol** is a set of rules that defines how data is transmitted and received between devices in a network.

Example:

Just as humans follow a language to communicate, computers follow protocols to exchange data.

1. TCP (Transmission Control Protocol)

Definition

TCP is a **connection-oriented and reliable** transport layer protocol.

Features

- Establishes connection before communication
- Provides error checking
- Guarantees packet delivery
- Uses acknowledgements (ACK)

Working

1. Connection established (3-way handshake)
2. Data transmitted
3. Receiver sends ACK
4. Lost packets are retransmitted

Applications

- Web Browsing
- Email
- File Transfer

Advantages

- Reliable
- Ordered delivery

Disadvantage

- Slower than UDP

Viva Question

Why is TCP reliable?

Answer: Because it uses acknowledgements, sequencing, and retransmission.

2. UDP (User Datagram Protocol)

Definition

UDP is a **connectionless transport layer protocol**.

Features

- No connection setup
- Fast transmission
- No acknowledgement
- No error recovery

Applications

- Video Streaming

- Online Gaming
- VoIP
- DNS

Advantages

- Fast
- Low overhead

Disadvantages

- Unreliable

Viva Question

Why is UDP faster than TCP?

Answer: UDP does not establish connections or send acknowledgements.

TCP vs UDP

TCP	UDP
Connection Oriented	Connectionless
Reliable	Unreliable
Slow	Fast
ACK Used	No ACK
Error Recovery	No Error Recovery

3. HTTP (HyperText Transfer Protocol)

Definition

HTTP is used for communication between web browsers and web servers.

Port Number

80

Uses

- Loading websites
- Sending web requests

Example

When you open a website, your browser sends an HTTP request.

Working

Browser → HTTP Request → Server

Server → HTTP Response → Browser

Viva Question

What is HTTP?

Answer: Protocol used to transfer web pages between browser and server.

4. HTTPS (HyperText Transfer Protocol Secure)

Definition

Secure version of HTTP.

Port Number

443

Features

- Uses SSL/TLS encryption
- Secure communication

Example

Online Banking

Online Shopping

Difference from HTTP

HTTP	HTTPS
Not Secure	Secure
Port 80	Port 443
No Encryption	Encrypted

5. FTP (File Transfer Protocol)

Definition

FTP is used for transferring files between computers.

Port Number

21

Uses

- Uploading website files
- Downloading files from server

Working

Client requests file transfer.

Server sends file.

Viva Question

What is FTP used for?

Answer: Transferring files between client and server.

6. DNS (Domain Name System)

Definition

DNS converts domain names into IP addresses.

Port Number

53

Example

google.com

becomes

142.250.x.x

Why Needed?

Computers understand IP addresses, not domain names.

Working

1. User enters google.com
2. DNS server receives request
3. Returns IP address

4. Browser connects to IP

Viva Question

What is DNS?

Answer: Protocol that converts domain names into IP addresses.

7. DHCP (Dynamic Host Configuration Protocol)

Definition

DHCP automatically assigns IP addresses to devices.

Ports

- Server → 67
- Client → 68

Information Assigned

- IP Address
- Subnet Mask
- Gateway
- DNS Server

DORA Process

D - Discover

Client searches DHCP server.

O - Offer

Server offers IP address.

R - Request

Client requests offered IP.

A - Acknowledge

Server confirms assignment.

Viva Question

What is DORA?

Answer:

Discover → Offer → Request → Acknowledge

8. ARP (Address Resolution Protocol)

Definition

ARP converts IP addresses into MAC addresses.

Example

Device wants to send data to:

IP = 192.168.1.10

It first finds corresponding MAC address.

Working

1. ARP Request broadcast
2. Target replies with MAC address
3. Data transmission starts

Viva Question

Why is ARP needed?

Answer: To find MAC address from IP address.

9. ICMP (Internet Control Message Protocol)

Definition

ICMP is used for error reporting and diagnostics.

Uses

- Ping
- Network troubleshooting

Example

ping google.com

Uses ICMP packets.

Viva Question

Which protocol does Ping use?

Answer: ICMP

10. SMTP (Simple Mail Transfer Protocol)

Definition

SMTP is used for sending emails.

Port Number

25

Working

Sender Mail Server → SMTP → Receiver Mail Server

Example

Gmail sending email to another user.

Viva Question

What is SMTP used for?

Answer: Sending emails.

11. POP3 (Post Office Protocol Version 3)

Definition

Used to receive emails from mail server.

Port Number

110

Feature

Downloads emails to local computer.

Viva Question

What is POP3?

Answer: Email receiving protocol.

12. IMAP (Internet Message Access Protocol)

Definition

Used for accessing emails directly on server.

Port Number

143

Advantage

Emails remain on server.

Example

Checking same Gmail account on mobile and laptop.

Difference Between POP3 and IMAP

POP3

Downloads Mail

Mail Removed From Server

Single Device

IMAP

Syncs Mail

Mail Stays On Server

Multiple Devices

13. Telnet

Definition

Remote login protocol.

Port Number

23

Feature

Allows remote access to another computer.

Problem

Not secure.

Replaced By

SSH

14. SSH (Secure Shell)

Definition

Secure remote login protocol.

Port Number

22

Features

- Encrypted communication
- Secure remote administration

Viva Question

Why SSH is preferred over Telnet?

Answer: SSH encrypts data while Telnet sends data in plain text.

Important Port Numbers (Very Frequently Asked)

Protocol Port

HTTP 80

HTTPS 443

FTP 21

SSH 22

Protocol Port

Telnet	23
SMTP	25
DNS	53
DHCP	67, 68
POP3	110
IMAP	143

10 Most Important Viva Questions**What is a protocol?**

Rules for communication between devices.

Which protocol is reliable?

TCP

Which protocol is faster?

UDP

Which protocol assigns IP addresses?

DHCP

Which protocol converts domain names into IP addresses?

DNS

Which protocol converts IP address to MAC address?

ARP

Which protocol is used by Ping?

ICMP

Which protocol transfers files?

FTP

Which protocol sends emails?

SMTP

Which protocol is used for secure web browsing?

HTTPS